Four days on the Kapton air holder

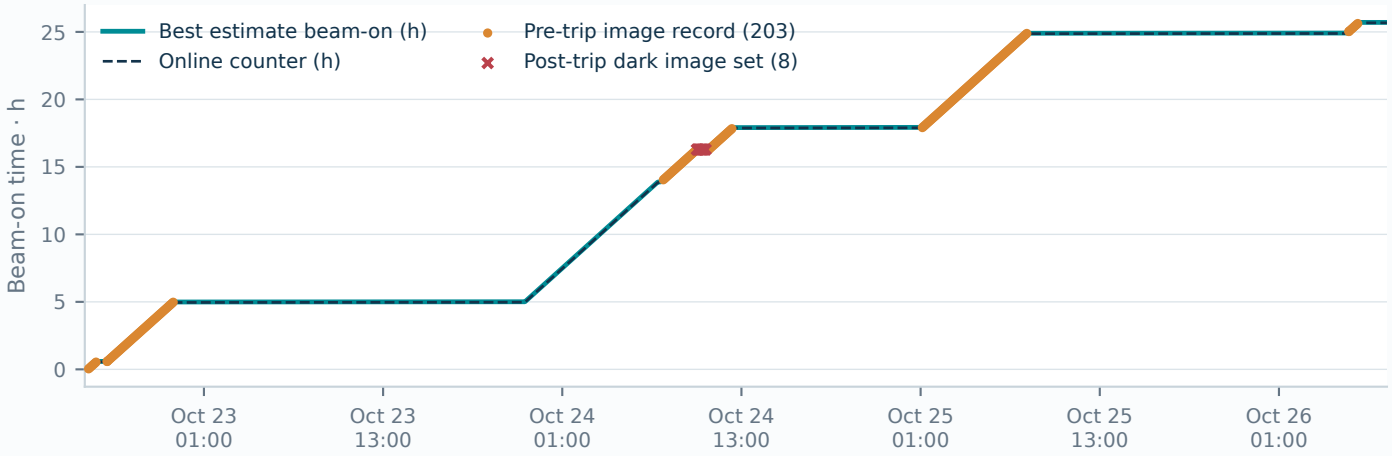
22–26 Oct 2026 | Nominal 6 keV | Post-repair ceiling: 0.009 Torr

25 h 42 min 17 s
ESTIMATED BEAM TIME
Online total + 116 s early correction

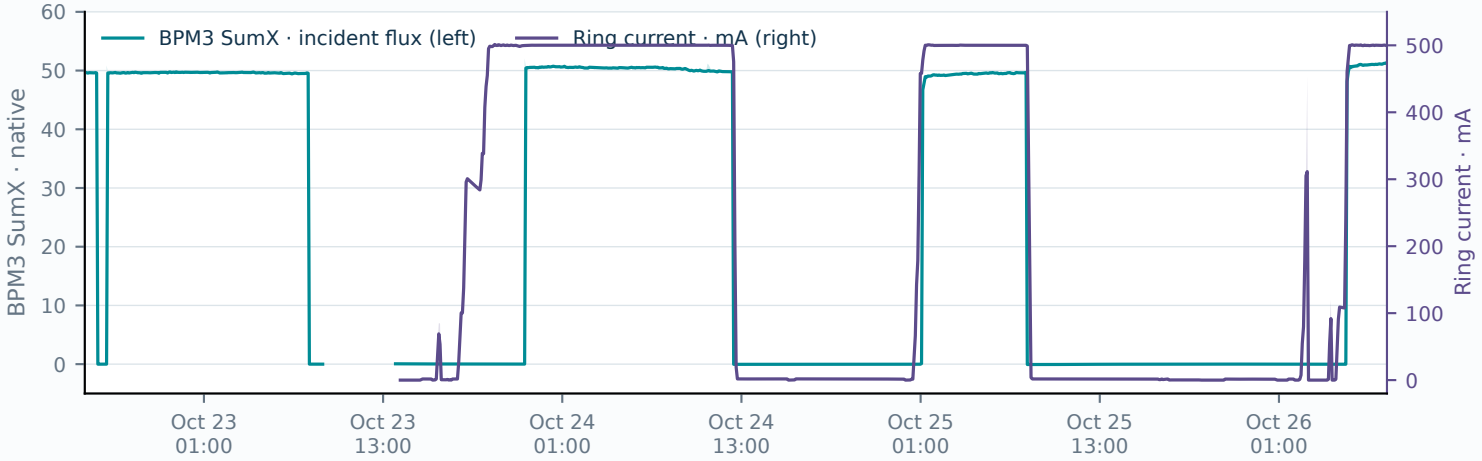
20 h 42 min 54 s
WEEKEND BEAM ADDED
Post-repair (Oct 23–26) at 2.6 mTorr

211 / 214 sets
JOURNAL END RECORDS
149 weekend sets; 8 after isolation

01 Cumulative beam-on exposure across five delivery episodes



02 Storage-ring current and incident X-ray flux (BPM3)



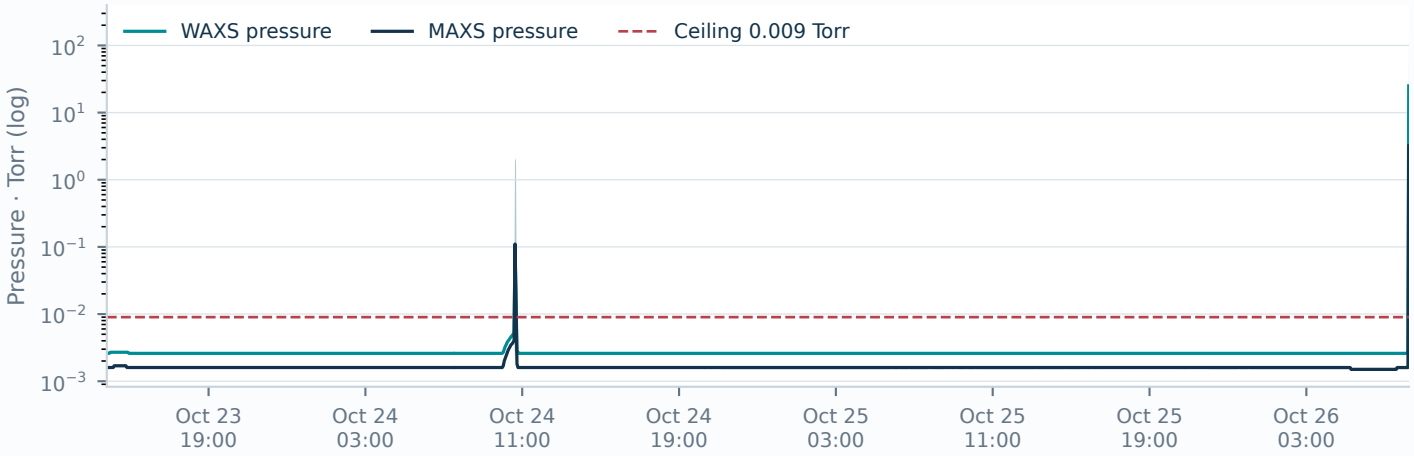
EXECUTIVE SUMMARY

• Log coverage: Oct 22 17:01 → Oct 26 08:14 EDT; 295,950 samples across 87 h 13 min.
• Post-repair beam-on pressures: WAXS 2.5–2.7 mTorr, MAXS 1.5–1.7 mTorr; SAXS zero denotes below range. No over-ceiling pressure was recorded during weekend beam-on samples.
• Saturday's stale-read trip completed staged shutdown; eight post-trip image end records followed. The pressure-reader fix exists on disk but is NOT evidenced in this database.
• Monday: last image attempt 06:21:45; first closed-shutter sample 06:21:47. A separate high-pressure trip at 08:13:15 completed shutdown by 08:13:48.
• Diode-confirmed photon and model-dose estimates use the operator's 40 × 300 µm footprint. Saturday first/last images and the dose derivation follow in this report.

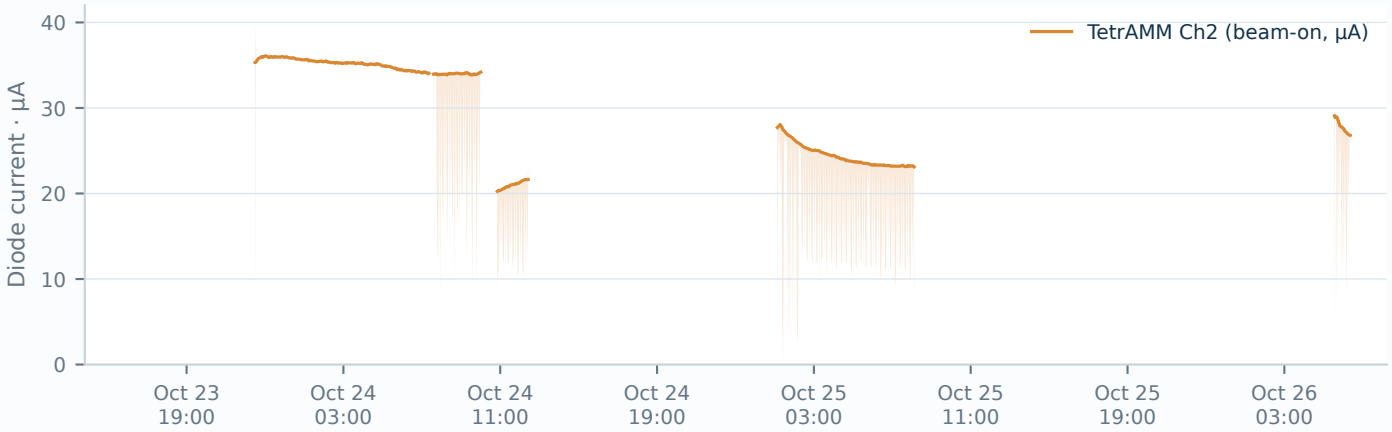
Post-repair vacuum & diode telemetry

0.009 Torr ceiling | Continuous TetrAMM channel 2 | Five irradiation episodes

03 Weekend chamber pressures (Oct 23 13:48 → Oct 26 08:14)



04 Transmitted diode current during beam-on intervals



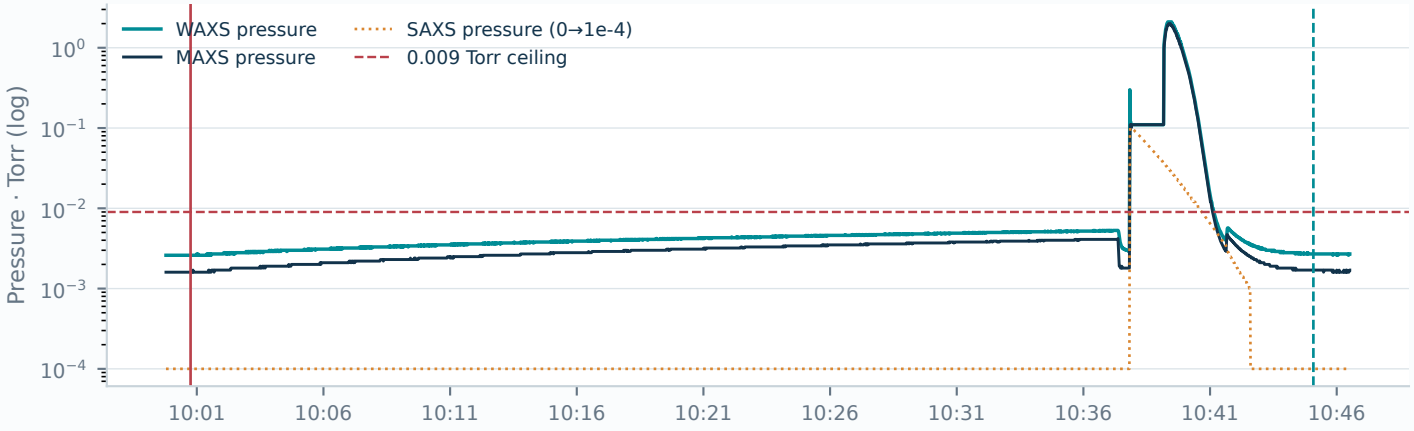
Episode / Window (EDT)	Beam Time	Cumul. Beam	BPM3 Mean	Diode (nA)	WAXS / MAXS
1: 10-22 17:01 → 10-22 22:57	4:59:22	4:59:22	49.64	n/a (pre-diode)	47.0 / 43.0 mTorr
2: 10-23 22:28 → 10-24 10:00	11:17:44	16:17:06	50.47	35,149	2.6 / 1.6 mTorr
3: 10-24 10:45 → 10-24 12:26	1:37:31	17:54:37	49.85	21,040	2.6 / 1.6 mTorr
4: 10-25 01:06 → 10-25 08:07	6:59:25	24:54:03	49.38	23,906	2.6 / 1.6 mTorr
5: 10-26 05:33 → 10-26 06:21	0:48:14	25:42:17	50.49	27,663	2.6 / 1.5 mTorr

Times in table: h:mm:ss. Diode medians select fresh beam-on samples >1000 nA; this display filter is not used for dose integration. Gaps are not interpolated. Signal changes alone do not establish Kapton damage or realignment. Episode windows omit a 0.10 s counter increment near Oct 26 02:55; the campaign total includes it.

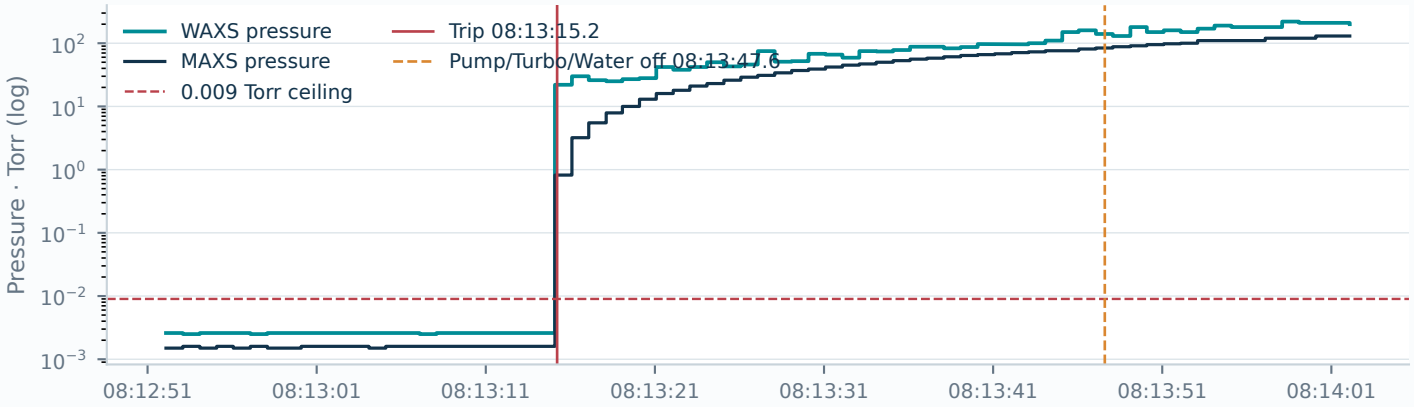
Protection trips & operational anomalies

SQLite evidence | Saturday stale-read trip | Monday pressure rise and image termination

05 Saturday: stale-read trip and recovery excursion



06 Monday: high-pressure trip after beam exposure ended



CHRONOLOGY OF ANOMALIES & SAFEGUARD ACTIONS

1. Saturday 10:00:45 — Stale-read trip. The notes identify a 3.87 s scalar-sweep delay exceeding the old 3 s freshness window; adjacent pressures were normal. Boundary readbacks confirmed at 10:00:48.526, IV1 at 10:01:19.327, turbo disable at 10:01:19.853, and water closure at 10:01:21.275. Eight post-trip image end records have zero added beam time. After a separate recovery excursion, the guard rearmed at 10:45:04.730.

2. Export backlog. The final journal reports cursor #234025 versus cutoff #289790: 55,765 pending scalar IDs. Oct 25 15:01:55.918 is the cursor's SOURCE timestamp, not when export stopped. Tiled scan 1185490 reports 101,417 scalar events; the backlog cause is not established.

3. Monday termination. The 06:21:44.845 image attempt has no end record; a close operation follows 34 s later. Tiled now confirms scan 1185490 failed: 'Image detector completion timed out'. The shutter reads closed at 06:21:47.284. At 08:13:15, WAXS/MAXS jumped to 22/0.82 Torr; shutdown readbacks completed at 08:13:47.603. The pressure-rise cause, including possible intentional venting, requires operator confirmation.

Valve confirmation uses 'Not Open' status, not an independent closed limit. Turbo OnOff=0 confirms disable, not rotor spin-down.

Campaign statistics & session audit

Integrated exposure counters, flux/diode distributions, and Bluesky single-run session records

Best-estimate beam time (campaign)	25 h 42 min 17 s (92,536.79 s)
Saved online beam counter (beam_s)	25 h 40 min 21 s (92,420.76 s)
Added Oct 22 early photon MINOR-alarm interval	116.03 s
Weekend beam time added (post-repair)	20 h 42 min 54 s (74,574.46 s)
Total wall-clock campaign span	87 h 12 min 48 s
Observed monitor time / Unobserved restart gaps	82 h 12 min 26 s / 5 h 00 min 22 s
Fast shutter controller-open time (fast_open_s)	63 h 10 min 45 s (227,445.38 s)
FE open + Fast shutter open time (fe_fast_open_s)	26 h 37 min 01 s (95,820.88 s)
BPM3 beam-weighted mean ± SD (overall)	49.973 ± 0.571 native (CV 1.14%)
BPM3 beam-weighted mean ± SD (weekend post-repair)	50.053 ± 0.596 native
Integrated BPM3 × beam time (overall)	4,624,364 signal·s
Filtered diode median (p10 – p90)	33,956 (23,176 – 35,716) nA
Beam-on medians: WAXS / MAXS / SAXS	0.0026 / 0.0016 / 0 Torr; SAXS below range
Image end records / attempts (SQLite)	211 / 214 (62/63 legacy + 149/151 single-run)
Pre-trip vs. Post-trip single-run image sets	141 pre-trip beam sets + 8 post-trip dark sets
Scalar / recovery / isolation records	295,950 / 20 / 17

SINGLE-RUN SESSIONS RECORDED IN SQLITE

End (EDT)	Start ID	Export ID	Cutoff ID	Pending	Image journal
Oct 23 13:57:19	57,284	57,543	57,666	123	Recovery failed; no beam
Oct 24 07:45:47	120,835	120,835	121,460	625	No image end records
Oct 24 10:37:29	121,467	131,756	131,758	2	27 pre-trip + 8 post-trip
Oct 24 10:46:22	132,252	132,289	132,292	3	Attempt #36 pending
Oct 24 10:51:35	132,459	132,600	132,604	4	One end record (#37)
Oct 26 06:31:22	132,608	234,025	289,790	55,765	113 end records; #151 pending

EVIDENCE & LIMITS
• Friday night delivered 8.88 h of logged beam time without image records in this journal.
• The database has no pressure_samples table or pressure_stale_s setting: deployment of Saturday's pressure-reader fix is not established.
• All image end records are flagged 'interrupted'; normal shutter cycling can set this flag. Selected first/last SAXS/WAXS/OAV arrays now resolve through Tiled.
• Source: checkpointed Oct 26 SQLite snapshot (path and full audit in JSON).
SHA-256: 7396e156d448cd138120ed1ec07461330f07aa84880549cc064afd3684d3854b

Initial optical & WAXS reference frames

Cached Oct 22 frames: scans 1185420 and 1185481 | Initial five-hour reference

EPISODE 1 START

Oct 22, 17:16:26 EDT (Oct 22)

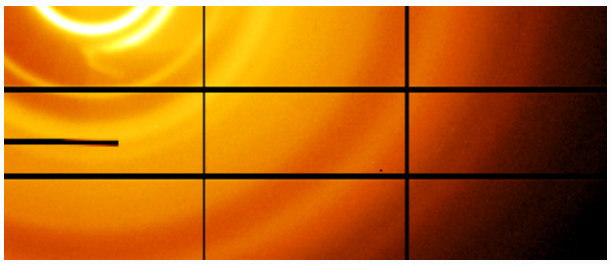
Beam on: 0 h 02 min 21 s

EPISODE 1 END

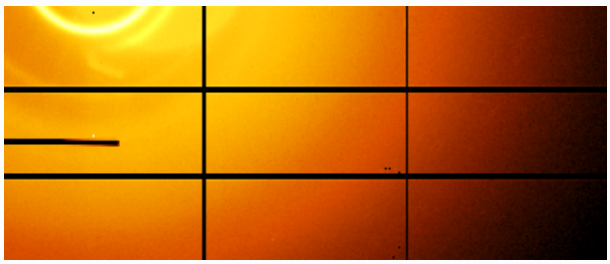
Oct 22, 22:56:27 EDT (Oct 22)

Beam on: 4 h 58 min 19 s

WAXS · scan 1185420



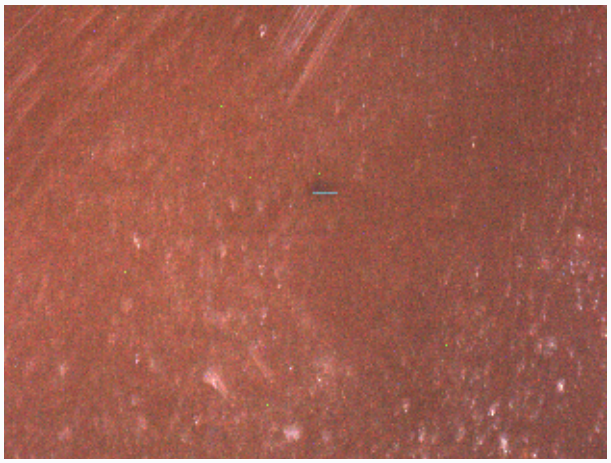
WAXS · scan 1185481



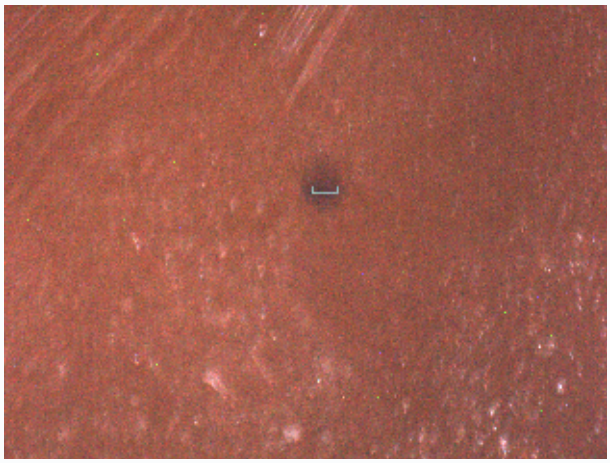


WAXS counts · fixed logarithmic yellow-hot scale

On-axis view (OAV)



On-axis view (OAV)



These are initial-campaign references, not final weekend images. The first frame already includes 2 min 21 s exposure. WAXS uses one fixed log scale with nonpositive pixels masked. The following Saturday comparisons use this exact scale; OAV retains its original RGB values.

Saturday: first low-pressure run

Scan 1185487 | First / last pre-trip images | Same WAXS count scale and original OAV RGB

BEFORE

Oct 24, 07:46:07 EDT

Event 1 | Campaign counter:

14 h 02 min 42 s

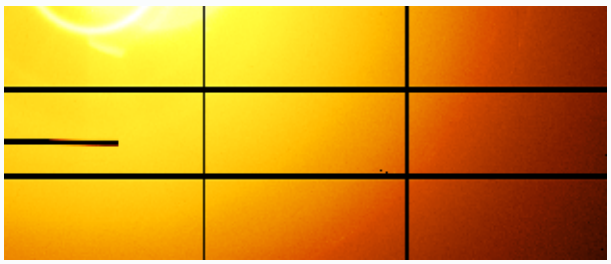
AFTER

Oct 24, 09:56:02 EDT

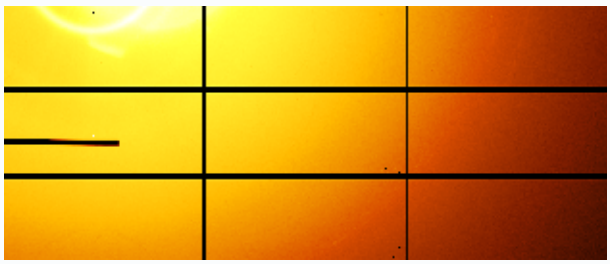
Event 27 | Campaign counter:

16 h 12 min 24 s

WAXS | 1 s | arc 15°



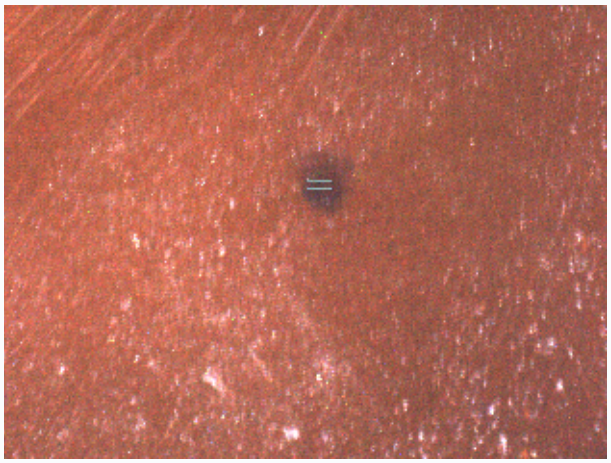
WAXS | 1 s | arc 15°



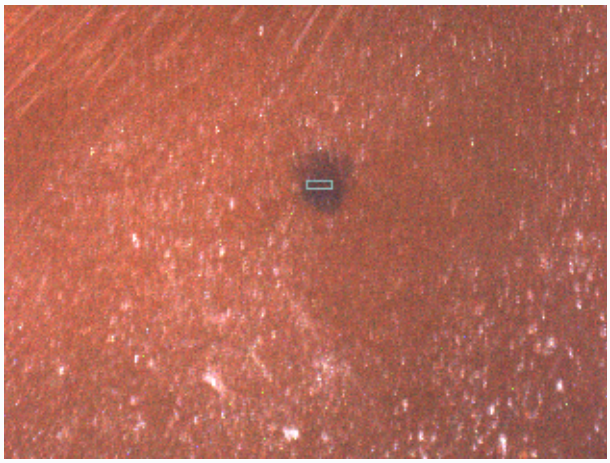


Fixed log counts: 496–18,068.5 (same as October 22)

On-axis view | original RGB



On-axis view | original RGB



Beam-counter increment between displayed frames: 2 h 09 min 41 s. Both exposures resolve through Tiled. 'Before' is the first image in this run, not virgin Kapton. Counter labels are campaign-wide and do not imply all prior exposure reached this position. No per-frame intensity normalization, color rescaling or image registration is applied. Nonpositive WAXS pixels are masked; bright values above the fixed scale saturate visually. Last pre-trip event 27 is used; events 28–35 were taken after boundary isolation.

Tiled UID: fca35755-e42b-4d04-9d0b-8d2c57e96702

Saturday–Monday: second long session

Scan 1185490 | First / last pre-trip images | Same WAXS count scale and original OAV RGB

BEFORE

Oct 24, 10:51:45 EDT

Event 1 | Campaign counter:

16 h 19 min 59 s

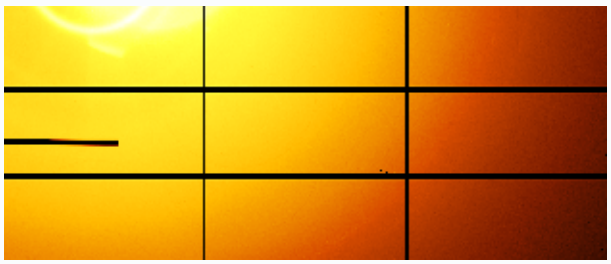
AFTER

Oct 26, 06:16:42 EDT

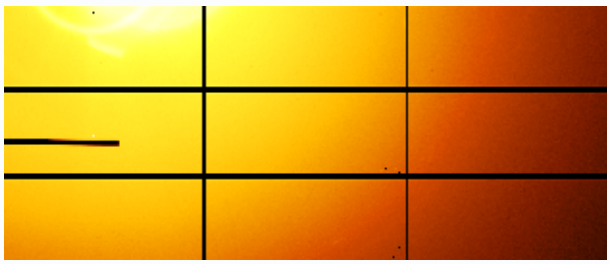
Event 113 | Campaign counter:

25 h 37 min 17 s

WAXS | 1 s | arc 15°



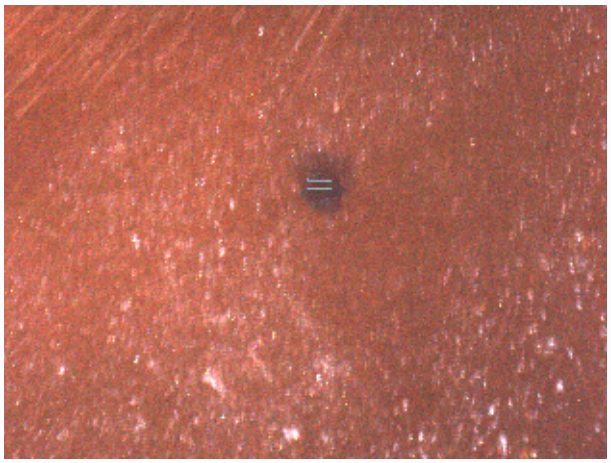
WAXS | 1 s | arc 15°



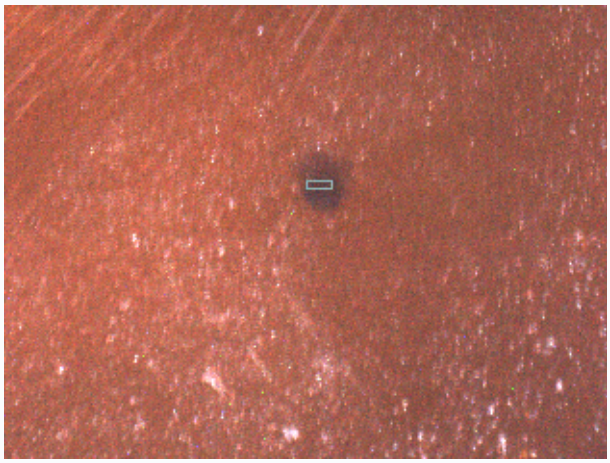


Fixed log counts: 496–18,068.5 (same as October 22)

On-axis view | original RGB



On-axis view | original RGB



Beam-counter increment between displayed frames: 9 h 17 min 17 s. Both exposures resolve through Tiled. 'Before' is the first image in this run, not virgin Kapton. Counter labels are campaign-wide and do not imply all prior exposure reached this position. No per-frame intensity normalization, color rescaling or image registration is applied. Nonpositive WAXS pixels are masked; bright values above the fixed scale saturate visually. Event 113 is the final completed set; the later 06:21 attempt timed out and is excluded.

Tiled UID: 3b958206-68d7-4659-9dac-164624aa3911

Diode photons & Kapton dose

6 keV | 110 µm Al downstream | Two 127 µm Kapton sheets | 40 × 300 µm bounding area

Selected exposure	Charge (C)	Transmitted photons	Two-sheet mean (Gy)	Rear-sheet mean (Gy)
All diode exposure*	2.2092	2.537e+17	3.817e+10	3.314e+10
Sat morning 1185487	0.2677	3.075e+16	4.626e+09	4.016e+09
Sat–Mon 1185490	0.8005	9.193e+16	1.383e+10	1.201e+10
Between 1185490 images	0.7935	9.112e+16	1.371e+10	1.190e+10

* All-diode photons may be summed, but the all-exposure dose is an equivalent single-footprint value across moved positions. Use the Sat–Mon run for a conservative post-restart spot dose. Prior exposure is not assigned to that spot.

Al transmission = 0.03261 (3.26%); two-sheet Kapton transmission = 0.58849 (58.85%). Illuminated mass = 4.3282 µg. The rear sheet receives less energy than the front; both are averaged over the full bounding rectangle.

Logged units: nA (CurrentScale2=10^9, operator-confirmed). Subtract 3.5 nA, the upper settled-dark envelope, rather than adding back the negative dark median. Integrate only fresh, confirmed-beam samples; use the smaller endpoint current and reject transitions/gaps.

Unity Si absorption/collection and 3.6 eV per electron–hole pair give the minimum photon count for a passive PIN diode. No correction for incomplete diode absorption, collection losses or extra downstream attenuation is applied.

These are conservative sampled model estimates under the stated geometry and local-deposition assumption, not a statistical confidence limit or a minimum at every point. The measured outer footprint lowers the area-average estimate; the beam core can receive more.

Dose calculation & traceability

Primary-beam transport with separate total and energy-absorption coefficients

$$I[A] = (\text{logged_current}[nA] - \text{dark_upper}[nA]) \times 10^{-9}$$

$$Q = \Sigma\, I_\text{lower} \times \Delta t_\text{accepted}$$

$$N_\text{diode} \geq (Q/e) \times (3.6\ \text{eV} / 6000\ \text{eV})$$

$$N_\text{after_Kapton} \geq N_\text{diode} / T_\text{Al}$$

$$N_\text{incident} \geq N_\text{after_Kapton} / T_\text{Kapton}$$

$$E_\text{abs} = N_\text{incident} \times E_\text{photon} \times (\mu_\text{en}/\mu) \times (1 - T_\text{Kapton})$$

$$D_\text{two_sheets}\ [Gy] = E_\text{abs}\ [J] / (\rho \times A \times 2t)\ [kg]$$

NIST at exactly 6 keV: Al $\mu/\rho = 115.300\ cm^2/g$; Kapton mixture $\mu/\rho = 14.699826$, $\mu_en/\rho = 14.239971\ cm^2/g$. Kapton density $1.42\ g/cm^3$; Al $2.699\ g/cm^3$. Weight fractions: H 0.026362, C 0.691133, N 0.073270, O 0.209235.

Scan 1185490: 9.2397 h retained by the conservative integration, from 9.3727 h online beam time. Q = 0.800512 C; diode photons ≥ 2.9978e+15; upstream incident photons ≥ 1.5621e+17; deposited energy in both sheets ≈ 59.8629 J. Front-sheet mean 1.5654e+10 Gy, rear-sheet mean 1.2008e+10 Gy. Exit-plane area mean 1.0487e+10 Gy (model depth-minimum, not a lateral pointwise minimum).

Sources: NIST X-ray mass coefficient tables (H, C, N, O, Al); NIST Kapton composition; LBNL X-ray Data Booklet §4.5; EPICS quadEM current-scale documentation. Full URLs, constants, charge totals and selection method are stored in the JSON. No flux is inferred for Oct 22 before diode logging or for restart gaps. Dose scales inversely with assumed illuminated area.